

# Desigualdad maximal de Kolmogorov

**Teorema 1 (desigualdad maximal de Kolmogorov).** Sean  $\{X_j\}_{j \in \mathbb{N}_n}$  variables aleatorias independientes y centradas (i.e.  $\forall j \in \mathbb{N} : \mathbb{E}[X_j] = 0$ ) y  $\forall j \in \mathbb{N}_n : X_j \in \mathcal{L}^2(\mathbb{P})$

$$\implies \forall t \in \mathbb{R} : \mathbb{P} \left( \max_{1 \leq k \leq n} |S_k| > t \right) \leq \frac{\mathbb{V}(S_n)}{t^2} \quad \text{donde} \quad S_k = \sum_{j=1}^k X_j.$$

**Demostración:** Definimos  $\forall k \in \mathbb{N}_n, t \in \mathbb{R}_+$  :

$$A_k(t) = \{\omega \in \Omega : |S_k(\omega)| > t \wedge \forall j < k : |S_j(\omega)| \leq t\}.$$

Entonces,  $\left\{ \omega \in \Omega : \max_{1 \leq k \leq n} |S_k(\omega)| > t \right\} = \bigsqcup_{k=1}^n A_k(t)$  y, por tanto,

$$\begin{aligned} \mathbb{V}(S_n) &= \mathbb{E}[S_n^2] = \int_{\Omega} S_n^2(\omega) d\mathbb{P}(\omega) \geq \int_{\{\max_{1 \leq k \leq n} |S_n| > t\}} S_n^2(\omega) d\mathbb{P}(\omega) = \sum_{k=1}^n \int_{A_k(t)} S_n^2(\omega) d\mathbb{P}(\omega) = \\ &\geq \sum_{k=1}^n \int_{A_k(t)} (S_n - S_k + S_k)^2 d\mathbb{P} = \sum_{k=1}^n \int_{A_k(t)} S_k^2 + 2S_k(S_n - S_k) + (S_n - S_k)^2 d\mathbb{P} \\ &\geq \sum_{k=1}^n \int_{A_k(t)} S_k^2 d\mathbb{P} + \sum_{k=1}^n 2 \int_{\Omega} \underbrace{\mathbb{1}_{A_k(t)} \cdot S_k}_{\rightarrow 0 \text{ (*)}} (S_n - S_k) d\mathbb{P} \\ &\geq \sum_{k=1}^n t^2 \cdot \mathbb{P}(A_k(t)) = t^2 \cdot \mathbb{P} \left( \max_{1 \leq k \leq n} S_k \right) \end{aligned}$$

(\*) Tenemos que  $\mathbb{1}_{A_k(t)} \cdot S_k$  es  $\sigma(X_1, \dots, X_k)$ -medible y, además,  $S_n - S_k = \sum_{j=k+1}^n X_j$  es  $\sigma(X_{k+1}, \dots, X_n)$ -medible. Por el teorema  $\Pi$ - $\lambda$ ,  $\mathbb{1}_{A_k} S_k$  y  $S_n - S_k$  son independientes

$$\implies \mathbb{E}[\mathbb{1}_{A_k(t)} S_k \cdot (S_n - S_k)] = \mathbb{E}[\mathbb{1}_{A_k(t)} S_k] \cdot \mathbb{E}[S_n - S_k] = 0. \quad \blacksquare$$

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